

# Ergo 3: Virtual Globetrotter

## Installation Guide

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# Requirements

## Setup

For running this project, it is recommended to have **two laptops and an immersive room**:

1. The first laptop will be used to run the webservice. Since the AI models will be running on the laptop itself, it is recommended to have a good **NVIDIA GPU**. For fast image generation and request processing, we suggest having **8GB of VRAM**. This laptop will also be used for storing all the images, so sufficient memory is required. We use ASUS TUF GAMING A16 with NVidia GeForce RTX 4070 Laptop GPU for this purpose.
2. The second laptop will be used to run the Elastic Search container. If you host the application on a more powerful machine, it is possible to run both webservice and the Elastic Search container on a single computer. The only requirement for this laptop is that it is able to run Docker Desktop application and Docker containers. More information can be found on [the official Docker website](#).
3. Immersive room's screens should be configured in sequential order. We did not configure anything in the immersive room ourselves. If you have any questions regarding the immersive room in The Core in Howest, please contact [HIT Lab](#).
4. The website can be accessed from any device connected to the internet. We used an **iPad tablet** as it is easy to use for the target audience. Accessing the website from another device not connected to the setup is not a requirement, but it will make the user engagement easier. **Use the website only on a single device to prevent the API from overloading.**

## Software

- We recommend to run the code and install all the necessary requirements with **Python 3.11** as newer versions of Python do not have the necessary support for some packages.
- For running frontend, which was made in Vue.js, **Node.js** is needed.
- For IDE, we recommend using **Visual Studio Code**.
- For Docker, we recommend downloading and installing **Docker Desktop**

The guide was made with the assumption that the **source code** is already downloaded on the user's machine. Otherwise, it can be found on our [GitHub](#).

# Installation Guide

## 1. Elastic Search

1. Open `ERGO3\_webservice/test\_backend` directory in VS Code.
2. Run `start\_docker.py`. This will use the `compose.yaml` file. The Elastic Search image will be installed automatically by running the code for the first time.

## 2. Webservice: API

1. Go to [Hugging Face](#) and create your account.
2. On the same website, go to *Access Tokens* page and create a new access token.
3. Open `ERGO3\_webservice/test\_backend` directory in VS Code.
4. Create a new file named `Constants.py`. Change the *<your token here>* to the access token you have just created and place the following code in that file:

```
HUGGINGFACE_API_KEY = "<your token here>"
```
5. Check if Python 3.11 is installed by entering the following command in the terminal:

```
py -0
```
6. If you do not see the 3.11 version installed, download it from [the official Python website](#).
7. Enter this command in the terminal in VS Code to create a new virtual environment. **Make sure you are in the directory `test\_backend`:**

```
py -3.11 -m venv .venv
```
8. Activate the virtual environment by entering the following command in the terminal:

```
.\.venv\Scripts\Activate.ps1
```
9. Download all the necessary packages for running the application:

```
pip install -r requirements.txt
```
10. Change directory to the `test\_backend/models` by entering the following command in the terminal

```
cd ./models
```
11. Download Real ESRGAN upscaling model by following the guide in the [official GitHub repository](#). You will need to follow the guide for the [Python script](#).
12. Download PyTorch (CUDA 11) by following [the guide from the official website](#).
13. Download CyberRealistic (fp16) model from [here](#) and place it in `test\_backend\models` folder.
14. Change the current directory back to the `test\_backend` by running the pasting the following command twice in the terminal:

```
cd ..
```

15. Try running the `pregenerate\_images.py` file. You should encounter the following error:

```
Traceback (most recent call last): File
"d:\InstallationGuide\ERGO3_webservice\test_backend\pregenerate_images.py", line 1,
in <module> from EnvironmentGenerator import EnvironmentGenerator File
"d:\InstallationGuide\ERGO3_webservice\test_backend\EnvironmentGenerator.py", line
1, in <module> from basicsr.archs.rrdbnet_arch import RRDBNet File
"D:\InstallationGuide\ERGO3_webservice\test_backend\.venv\Lib\site-
packages\basicsr\__init__.py", line 4, in <module> from .data import * File
"D:\InstallationGuide\ERGO3_webservice\test_backend\.venv\Lib\site-
packages\basicsr\data\__init__.py", line 22, in <module> _dataset_modules =
[importlib.import_module(f'basicsr.data.{file_name}') for file_name in
dataset_filenames]
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"D:\InstallationGuide\ERGO3_webservice\test_backend\.venv\Lib\site-
packages\basicsr\data\__init__.py", line 22, in <listcomp> _dataset_modules =
[importlib.import_module(f'basicsr.data.{file_name}') for file_name in
dataset_filenames]
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Program
Files\WindowsApps\PythonSoftwareFoundation.Python.3.11_3.11.2544.0_x64__qbz5n
2kfra8p0\Lib\importlib\__init__.py", line 126, in import_module return
_bootstrap._gcd_import(name[level:], package, level)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File
"D:\InstallationGuide\ERGO3_webservice\test_backend\.venv\Lib\site-
packages\basicsr\data\realesrgan_dataset.py", line 11, in <module> from
basicsr.data.degradations import circular_lowpass_kernel, random_mixed_kernels File
"D:\InstallationGuide\ERGO3_webservice\test_backend\.venv\Lib\site-
packages\basicsr\data\degradations.py", line 8, in <module> from
torchvision.transforms.functional_tensor import
rgb_to_grayscaleModuleNotFoundError: No module named
'torchvision.transforms.functional_tensor'
```

16. Click on the path `[<your path>\ERGO3\\_webservice\test\\_backend\.venv\Lib\site-packages\basicsr\data\degradations.py](#)` and change the line

```
`from torchvision.transforms.functional_tensor import rgb_to_grayscale`
```

to line:

```
`from torchvision.transforms.functional import rgb_to_grayscale`
```

17. Save the file and try running the `pregenerate\_images.py` file again. You should not encounter any errors now.

### 3. Webservice: Frontend

1. Open the `ERGO3\_frontend` directory in VS Code.
2. Check if Node.js is installed by running the following command in the terminal:

```
node -v
```

3. Run the following command in the terminal to install all the packages needed to run the frontend:

```
npm install
```

### 4. Configuring Nginx

For port forwarding we used nginx. This allows us to have both frontend service and API service on a single link.

1. Install nginx from [the official website](#). Choose the drive `C:` root directory as a download path.
2. Open `C:\nginx-1.24.0\conf\nginx.conf` file in Notepad and change the content to the following configuration:

```

worker_processes 1;

events {
    worker_connections 1024;
}

http {
    include mime.types;
    default_type application/octet-stream;
    sendfile on;

    # Extend all timeout settings
    keepalive_timeout 3600;
    proxy_connect_timeout 3600;
    proxy_send_timeout 3600;
    proxy_read_timeout 3600;
    send_timeout 3600;
    fastcgi_read_timeout 3600;

    # Buffer settings
    proxy_buffer_size 128k;
    proxy_buffers 4 256k;
    proxy_busy_buffers_size 256k;

    server {
        listen 80;
        server_name localhost;

        # Frontend routing
        location / {
            proxy_pass http://localhost:5173;
            proxy_http_version 1.1;
            proxy_set_header Upgrade $http_upgrade;
            proxy_set_header Connection 'upgrade';

```

3. Save the file before closing the Notepad.

## 5. Configuring Ngrok

To avoid our website domain changing every time the application is run we used Ngrok service that allows to have a single static domain.

1. Download Ngrok from [the official website](#) by following the guide there.
2. Go to the [Home page](#) of the website and create your free account or log in if you have one already.
3. Navigate to the Domain page and create a new domain.
4. Open Command Prompt on your machine and enter the following command:

```
ngrok config edit
```

A configuration file will open on your screen immediately.

5. Change the configuration file:

```
version: "3"

agent:

  authtoken: <your token>

tunnels:

  web:

    proto: http

    addr: 80

    hostname: <your ngrok static domain address>
```

Change the `authtoken` to your token from Ngrok and `hostname` to your static domain address. Both of those can be found on the Ngrok website.

6. Save the file and exit the Notepad.

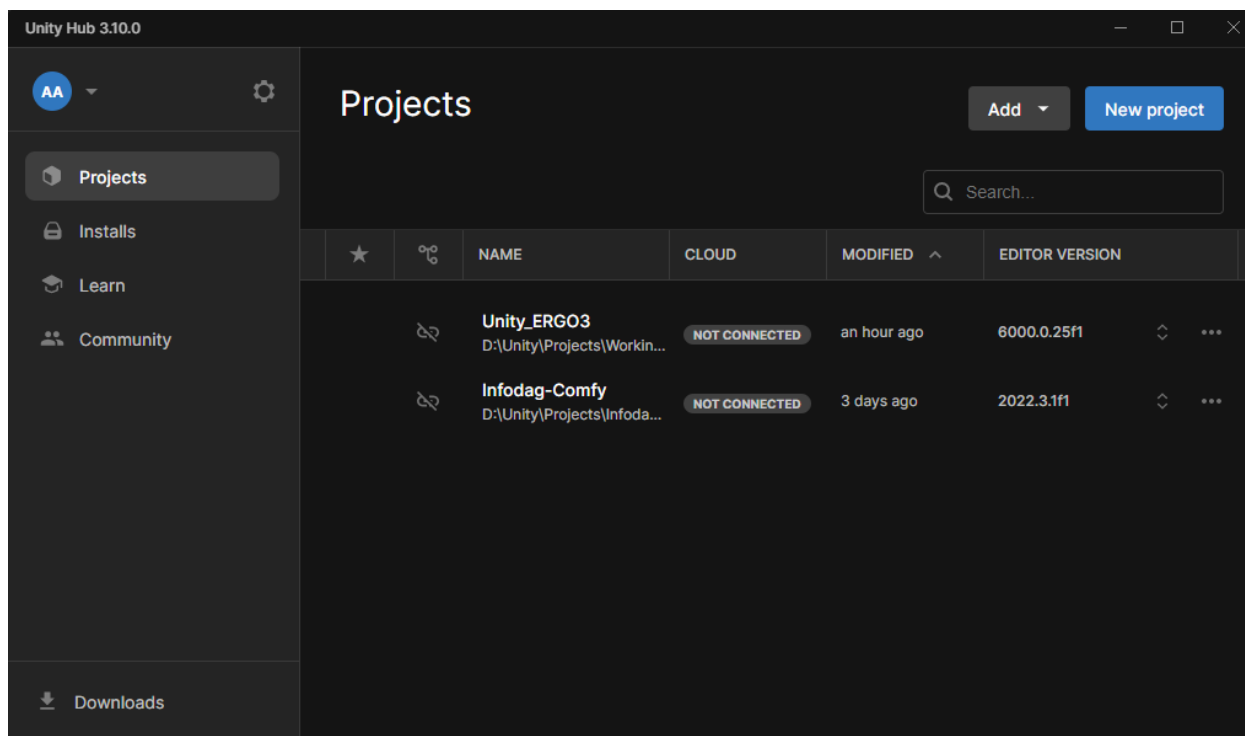
## 6. (Optional) Unity

To change either the Slideshow Images, or the Background Music, you need to open the project through **Unity**.

This project was developed using Unity version 6000.0.25f1, we recommend you install the same, or similar version.

1. To download Unity, you first need to [visit the website](#).

2. Follow the steps in the Install Wizard, and then open the Unity Hub, which should look similar to this:



3. Click on “Add”, then select “Add project from disk”.
4. A Windows File Explorer window will open. You now need to navigate to the folder “Unity\_ERGO3”. Click on it once (just select it, don’t open it), and press the “Add Project” button in the bottom-right corner.  
It should now show up in the “Projects” tab in Unity Hub. You can click on it to open the project in Unity.

To change images or music, please follow the steps in the chapter [Customization](#).

# Setup Guide

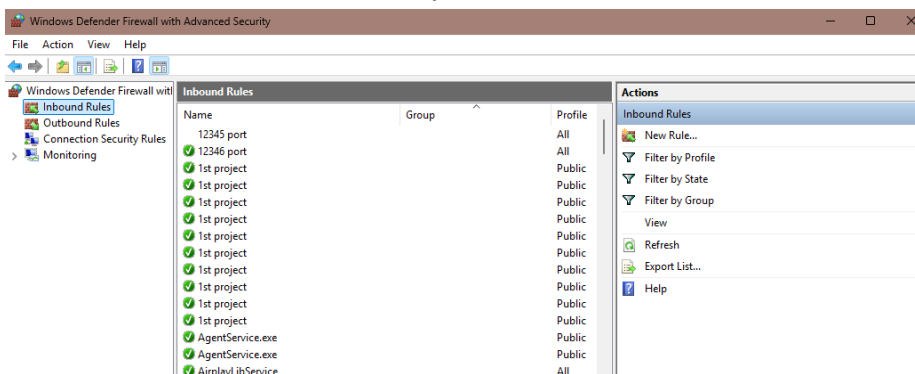
## STEP 1 (Optional): Configure Windows Defender Firewall

This step is optional, because it is already configured in the Howest immersive room.

1. Navigate to the Windows Start Menu.
2. Type “Windows Defender Firewall with Advanced Security” in the search box, and click on the first result:



3. Click on “Inbound Rules”, then press the “New Rule...” button



4. Select “Port”, then click “Next >”
5. Select “TCP”, and “Specific local ports”, then enter this: “12345, 12346” in the dialog box.
6. Click “Next >” until you have to name the rule.
7. Pick a memorable name (We used “12345 port” in this example), but you can name it however you want to.

## STEP 2: Run Elastic Search in a Docker container:

1. Start docker on the secondary laptop by running Docker Desktop.
2. Open `ERGO3\_webservice/test\_backend` directory in VS Code.
3. Activate the virtual environment by entering the following command in the terminal:  

```
.\venv\Scripts\Activate.ps1
```
4. Run the file `start\_docker.py`.



Make sure to have the `compose.yaml` file in the `test\_backend` folder.

## STEP 3: Run the Executable file in the Immersive room:

1. To run the project, navigate to the ERGO3\_Virtual\_Globetrotter folder, and double-click on the "Virtual Globetrotter.exe" file.

## STEP 4: Run the API code on your main laptop:

1. Open ERGO3\_webservice folder in Visual Studio Code.
2. Navigate to "ERGO3\_webservice/test\_backend" folder by running this command in the Visual Studio Code terminal:

```
cd ./test_backend
```

3. Activate the virtual environment by running the following command:

```
.\venv\Scripts\Activate.ps1
```

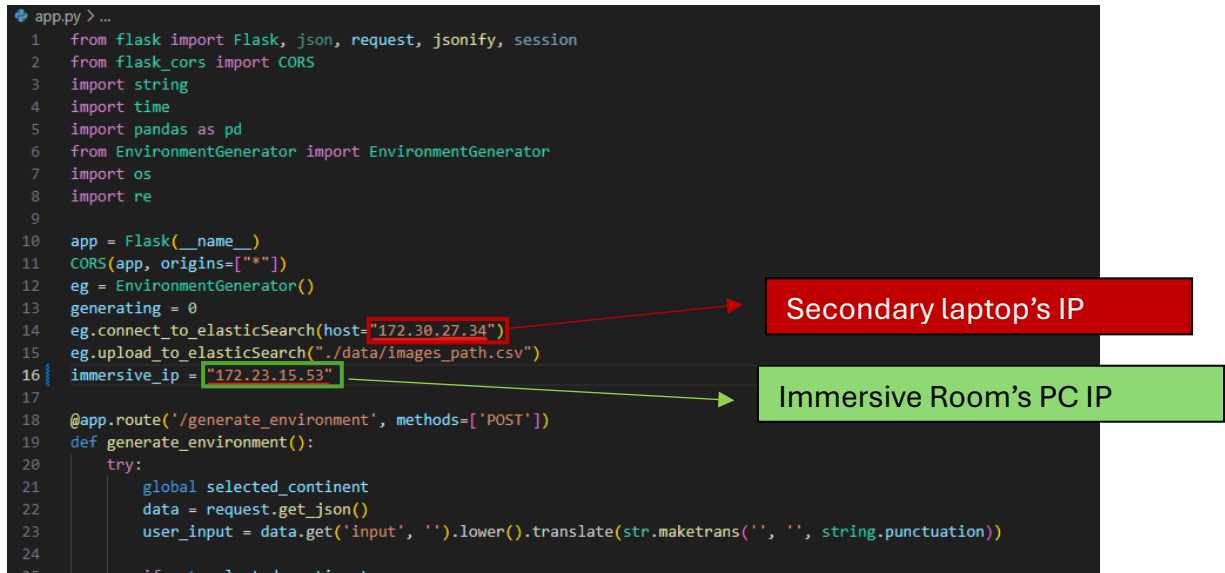
4. Retrieve the IP-addresses of the **secondary laptop and the immersive room PC**:
  - Open the Command Prompt and run the following command on both devices:  

```
ipconfig
```

- Copy the address under the name of IPv4 Address:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::bef3:e081:543b:8343%16  
IPv4 Address. . . . . : 192.168.68.119  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::da07:b6ff:fe65:abcd%16  
192.168.68.1
```

5. Open the “app.py” file on your main laptop and replace the following IP-addresses with the ones you retrieved.



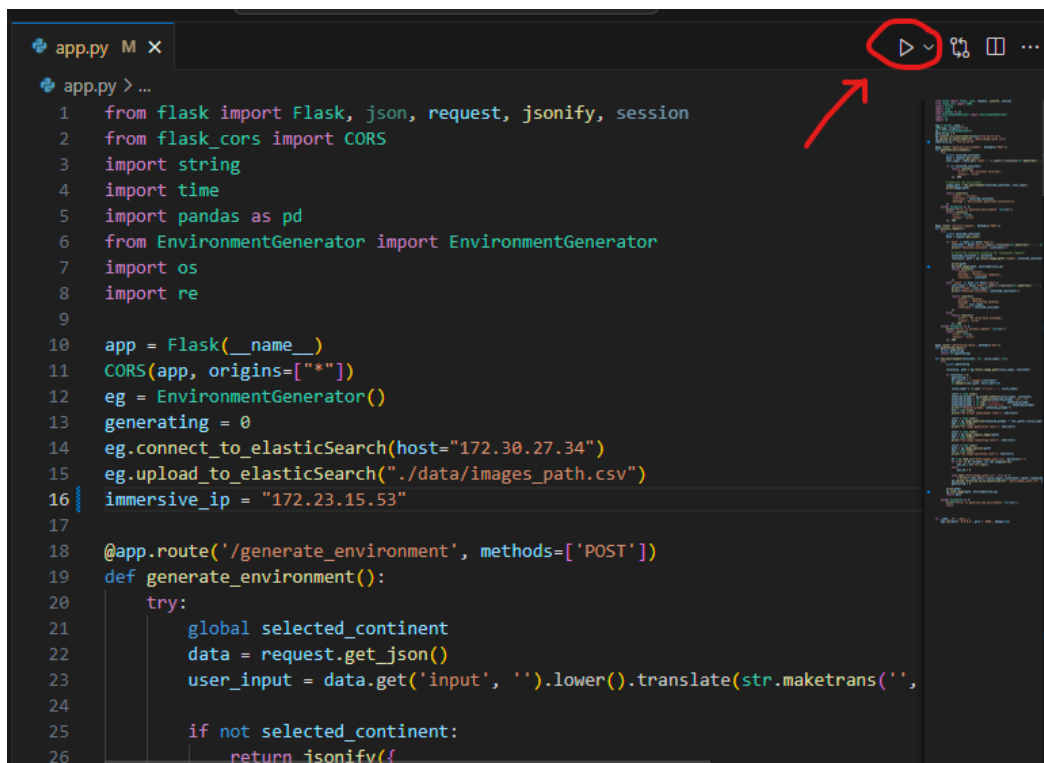
The screenshot shows a code editor with the following code in app.py:

```
1 from flask import Flask, json, request, jsonify, session  
2 from flask_cors import CORS  
3 import string  
4 import time  
5 import pandas as pd  
6 from EnvironmentGenerator import EnvironmentGenerator  
7 import os  
8 import re  
9  
10 app = Flask(__name__)  
11 CORS(app, origins=["*"])  
12 eg = EnvironmentGenerator()  
13 generating = 0  
14 eg.connect_to_elasticSearch(host="172.30.27.34")  
15 eg.upload_to_elasticSearch("./data/images_path.csv")  
16 immersive_ip = "172.23.15.53"  
17  
18 @app.route('/generate_environment', methods=['POST'])  
19 def generate_environment():  
20     try:  
21         global selected_continent  
22         data = request.get_json()  
23         user_input = data.get('input', '').lower().translate(str.maketrans('', '', string.punctuation))  
24  
25         if not selected_continent:
```

Two callouts are present:

- A red box labeled "Secondary laptop's IP" points to the host address "172.30.27.34" on line 14.
- A green box labeled "Immersive Room's PC IP" points to the immersive\_ip value "172.23.15.53" on line 16.

6. Run the “app.py” file and wait a minute for the all the packages to initialize.



The screenshot shows the same code editor with the app.py file. A red arrow points to the run button (a play icon) in the top right corner of the editor's toolbar.

## STEP 5: Running the frontend of the website:

1. Open the folder “ERGO3\_frontend” in Visual Studio Code.
2. Run the following command to start the web-application:

```
npm run dev
```

## STEP 6: Forwarding the webservice to a static domain:

1. Open the Command Prompt and run the following commands to start up the nginx:

```
cd C:\nginx-1.24.0
```

```
nginx start
```

2. Open the Command Prompt and run the following command to start up the ngrok:

```
ngrok start --all
```

Now the web application is available on the static domain that you have configured.

# Customization

## Choosing the right file

If you want to edit some parts of the Immersive Room experience, like the image Slideshow, or ambient music, then you have to open the project through Unity. To do this, follow the step ([Optional Install Unity in the previous chapters](#)).

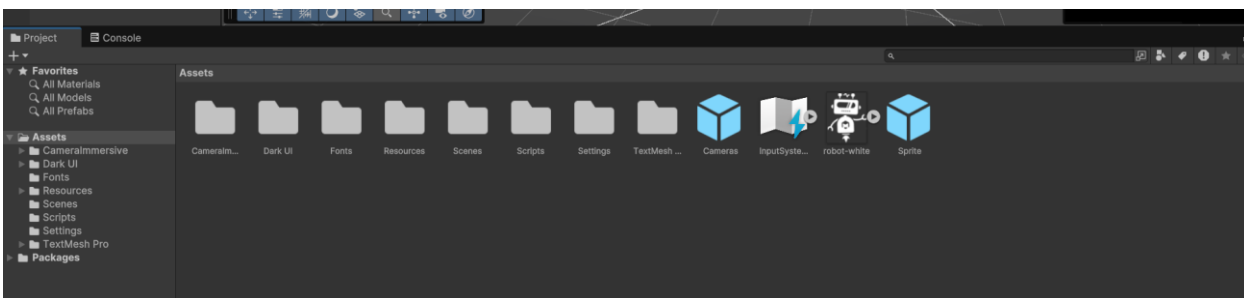
## Changing the image Slideshow, and Music

Editing the project images or music can be done with any computer (but preferably one with a good CPU).

If you wish to change the image Slideshow that is shown when the project is run, or the music that is played in the Immersive Room, you first need to open the Unity project file (Please, see previous step).

**If you make any changes to the project, make sure to build the project.**

Once you have done this, you should see the contents of the **Assets** folder on the bottom part of the screen. This folder contains everything that the project needs to run.

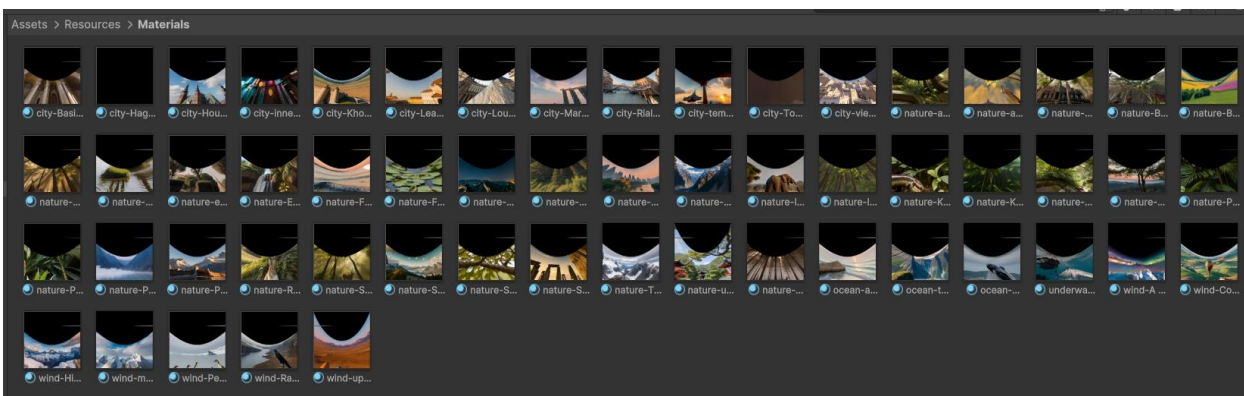


The images are located under the folder **Resources**.

## Deleting Slideshow images

To delete images shown in the slideshow, make sure that you are in the Resources folder. Click once on the image you want to delete and press the `Delete` key.

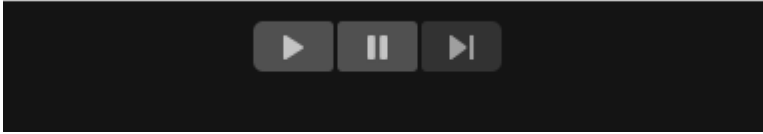
Then, navigate to the folder **Materials**, which is also located inside the Resources folder. You should see something similar to the picture below.



One file should have a grey picture. This is the material corresponding to the image you just deleted. Click once to select it, then press `Delete` again.

## Adding new Slideshow images

To add a new image, you first need to run the project through Unity. You can do this by pressing the `Play` button on the top side of the screen.



Rename the image you want to add to the project, by adding one of “city”, “nature”, “wind”, “ocean”, or “underwater”, followed by a dash (-), and the filename.

*Example: city-new\_york.jpg*

This will tell the program what kind of sound to play when the image is shown. If none of these prefixes are added, the audio is picked from the “Ambient” category.

Then, navigate to the Python script, and send an image to the port 12345. The new image should be displayed on the screen.

You can now stop the project from running, by pressing the `Stop` button, which should be in place of the `Play` button.

## Adding or deleting music

Navigate to the Resources folder. Take a look at the images present, and note that each image name is preceded by a **category name** (one of “city”, “nature”, “wind”, “ocean”, or “underwater”).

Now navigate to the **Audio** folder, which is located under Resources.

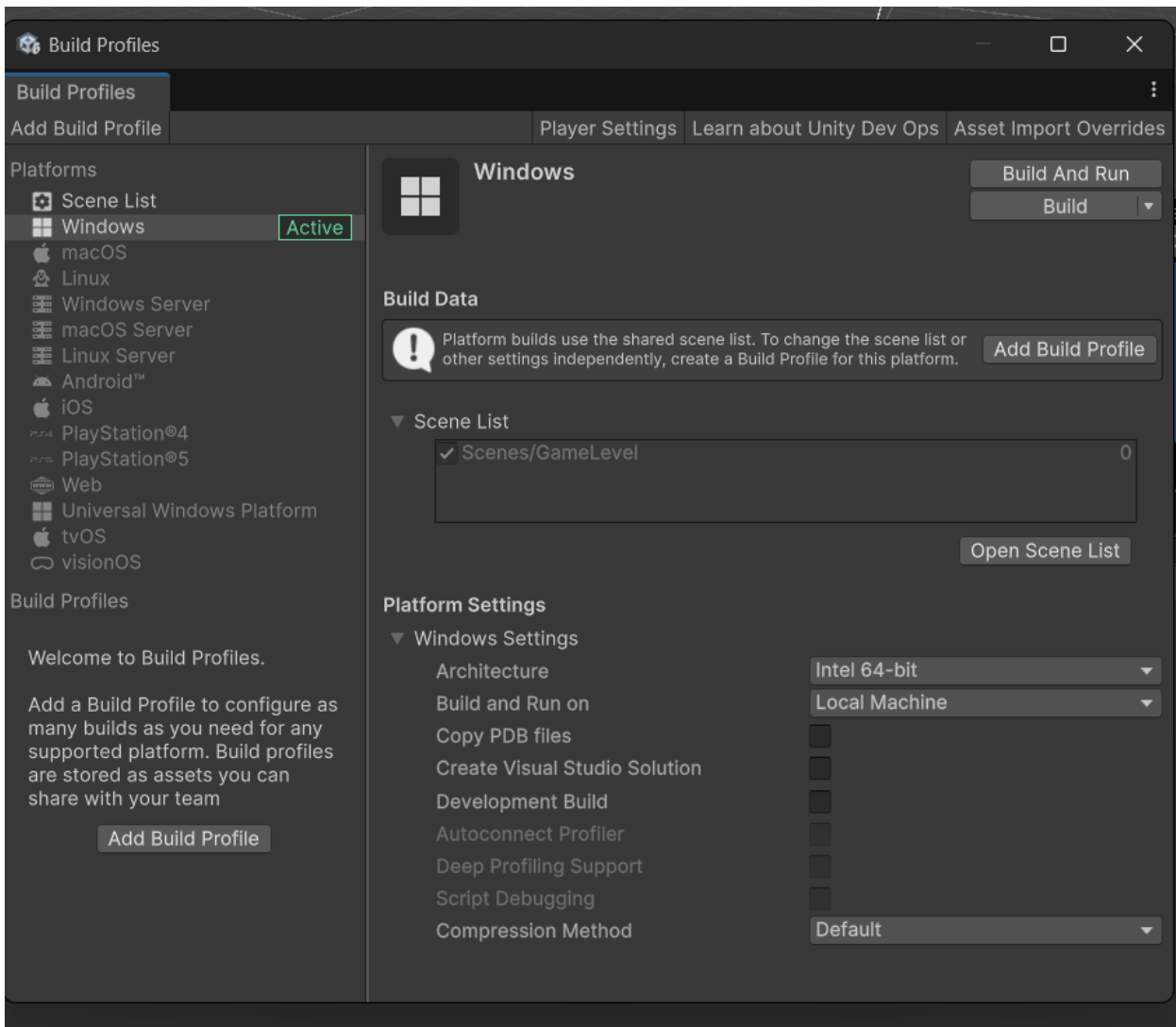
You will see 6 different folders. Each folder contains music that is played if the image currently displayed has the category in the start of its filename. The **one exception** to this is the **Ambient** category. Audio from this category will be played if the image name does not contain any category other than “city”, “nature”, “wind”, “ocean”, or “underwater”.

To add or delete music, simply navigate to the file you want to remove, click once to select it, and press the `Delete` key.

## Building a new project

To make your changes persist, you need to **rebuild** the project.

To do this, press the keys `Ctrl + Shift + B` which should open the **Build Profiles** window.



Then, press the button `Build` in the top right corner of this window.

A Windows File Explorer window should open, asking you to specify a Build folder. You can build the project anywhere, just make sure to note the folder you have chosen, as this is where the .exe file will be.